

Data Compression

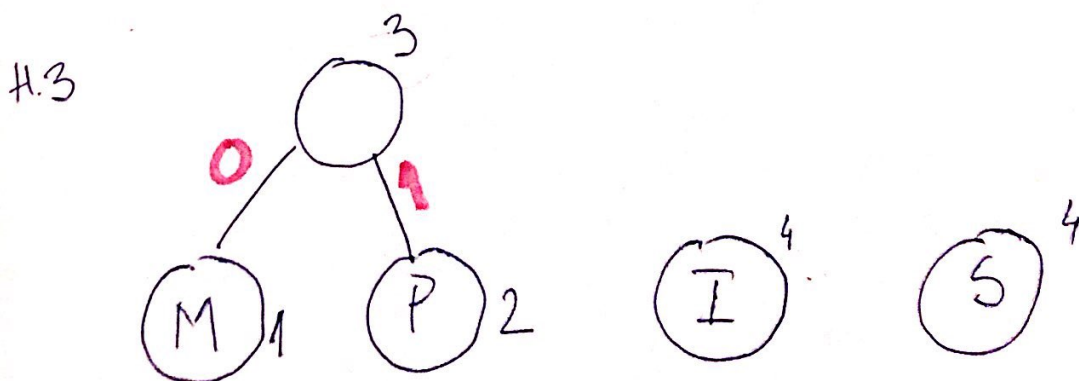
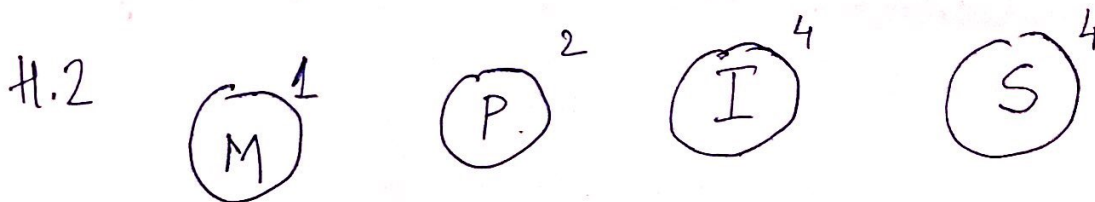
Huffman

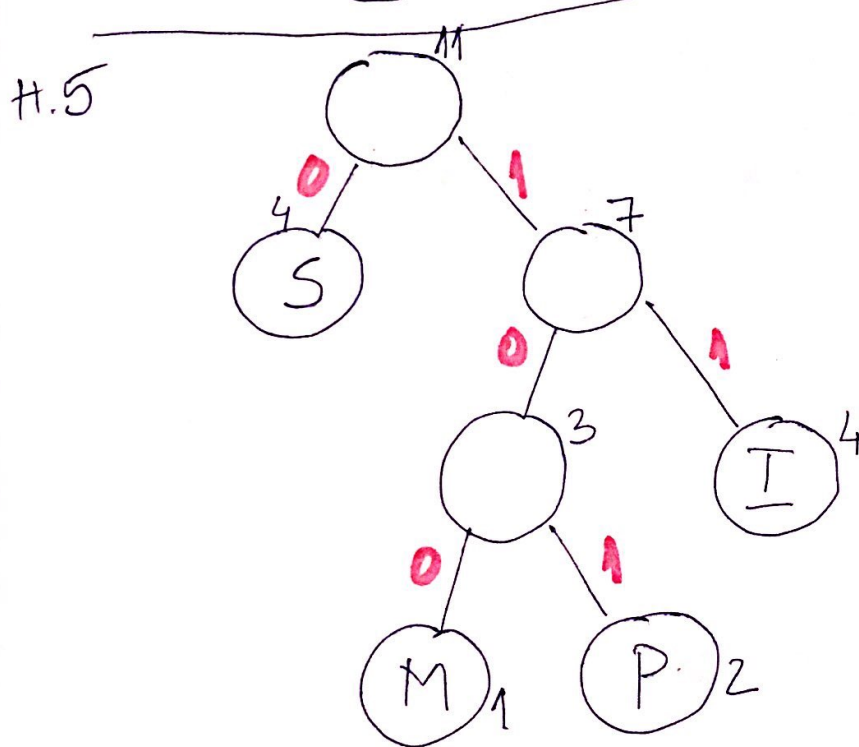
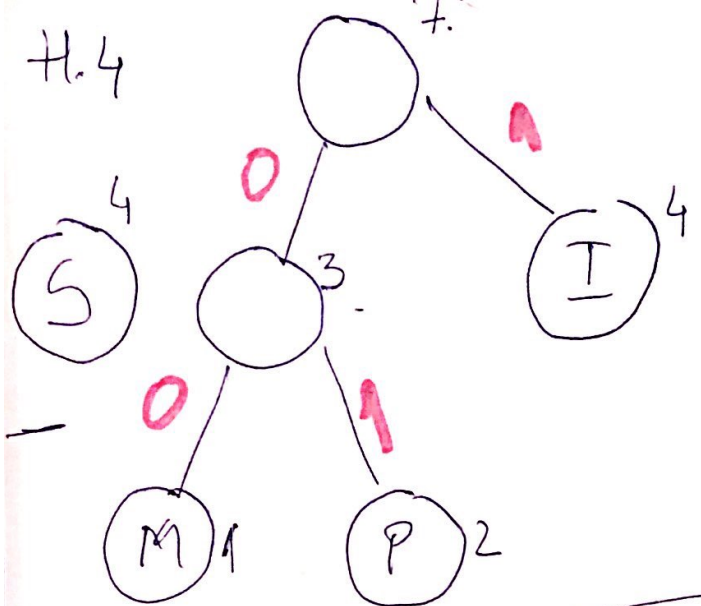
① Të kompresohet teksti i mëposhtëm duke përdorur Huffman Coding.

MISSISSIPPI

H.1

char	frequency	encoding
I	4	11
M	1	100
P	2	101
S	4	0





M = 100

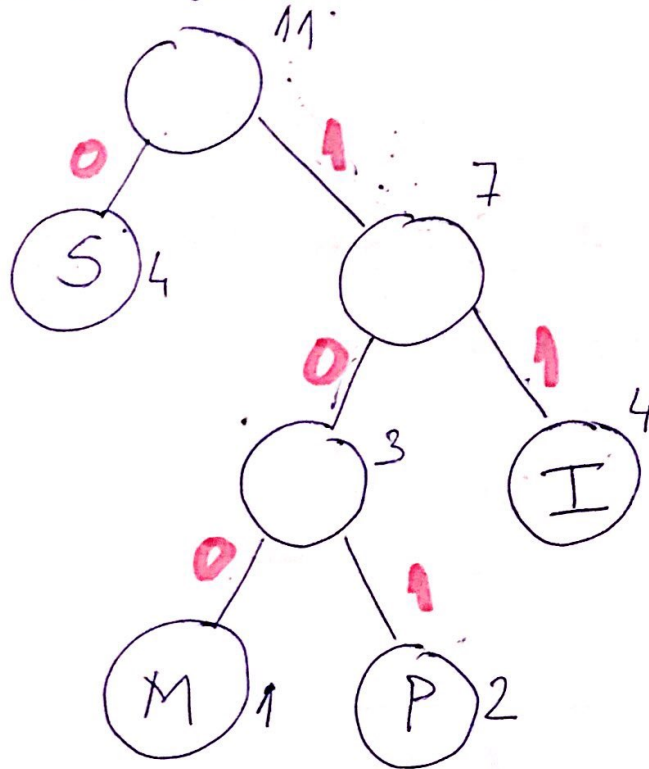
I = 11

P = 101

S = 0

MISSISSIPPI
 ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
 100 11 00 11 00 11 101 101 11

(2) Të dekompresohet teksti i mëposhtëm:
 mëse është dhënë truci



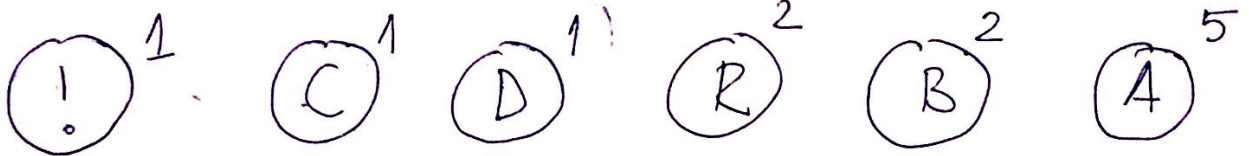
100 11 00 11 0 0 111 01 101 11
 ↓ ↓ ↓ ↓
 M I S S I S S I P P I

③ ABRACADABRA!

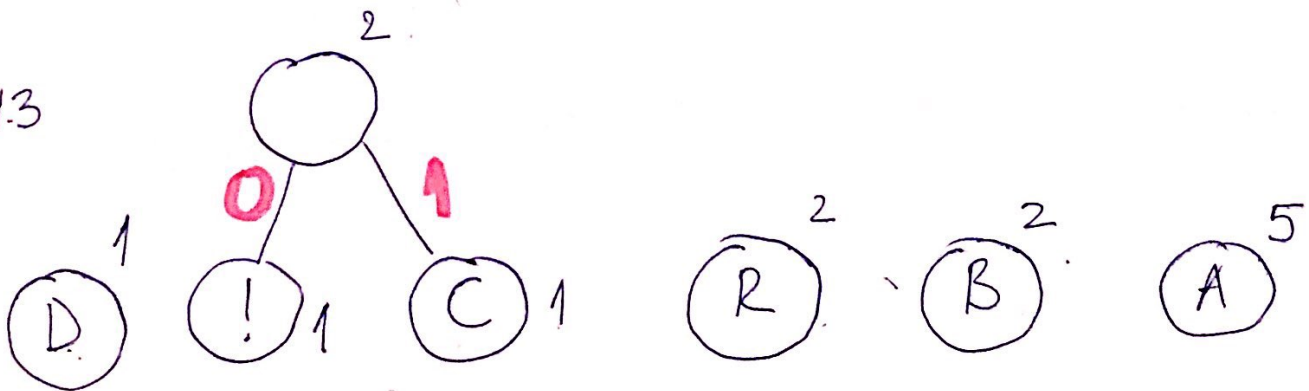
H.1

char	frequency	encoding
A	5	0
B	2	111
C	1	1011
D	1	100
R	2	110
!	1	1010

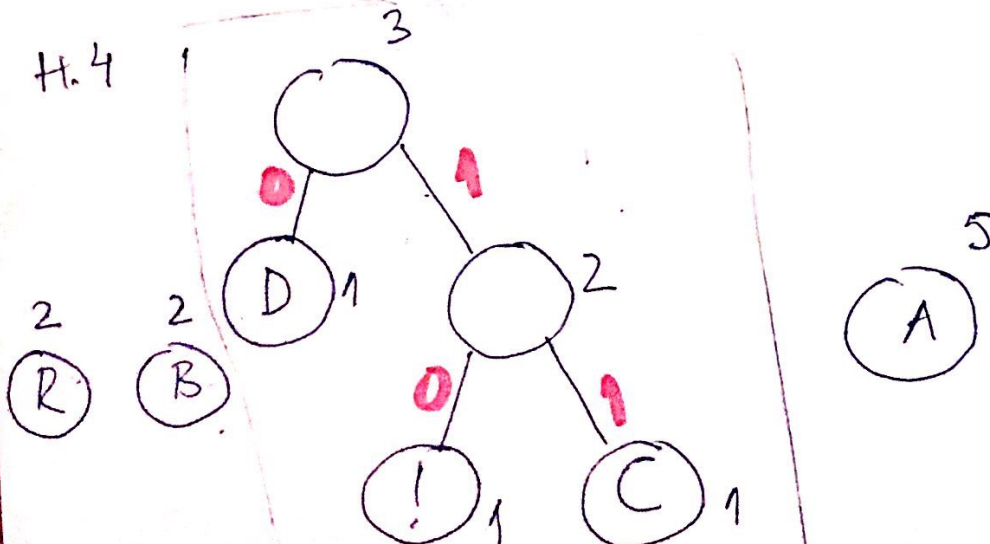
H.2

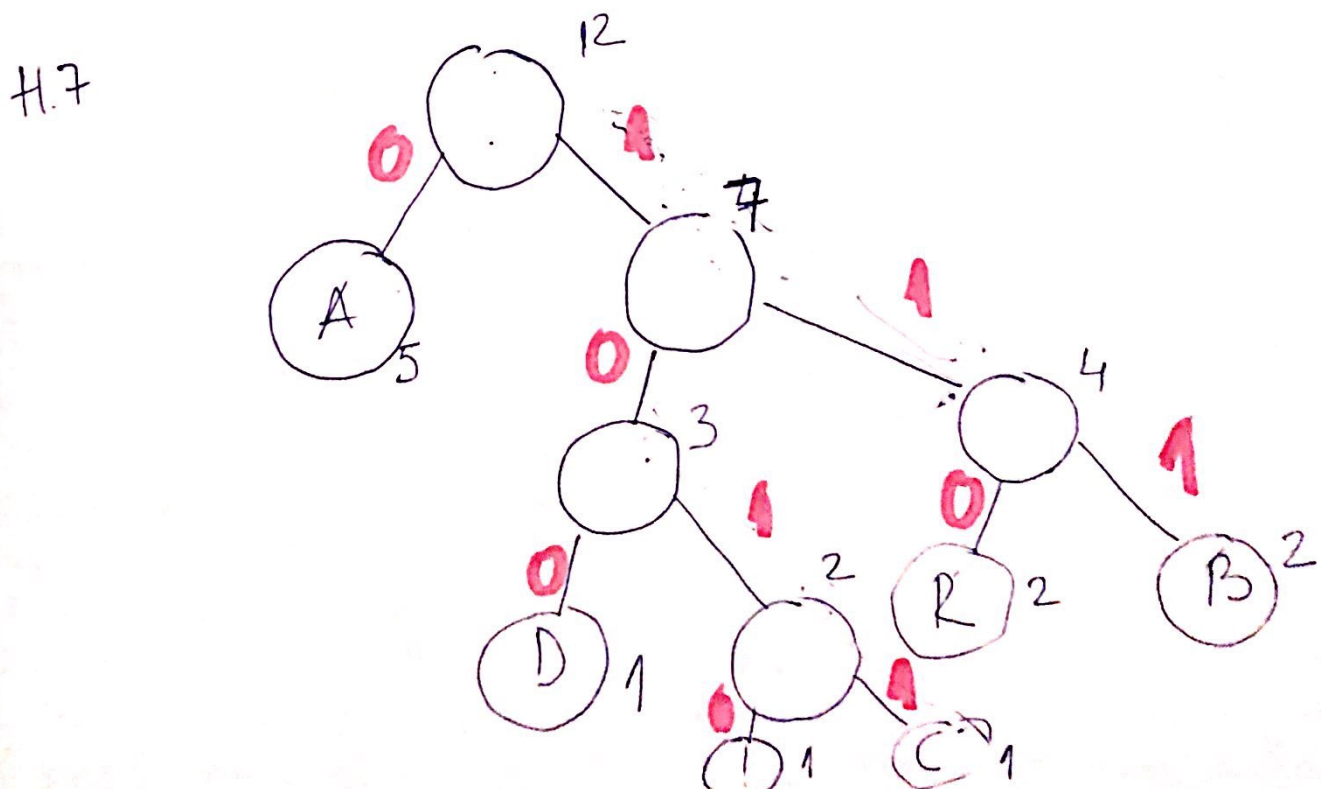
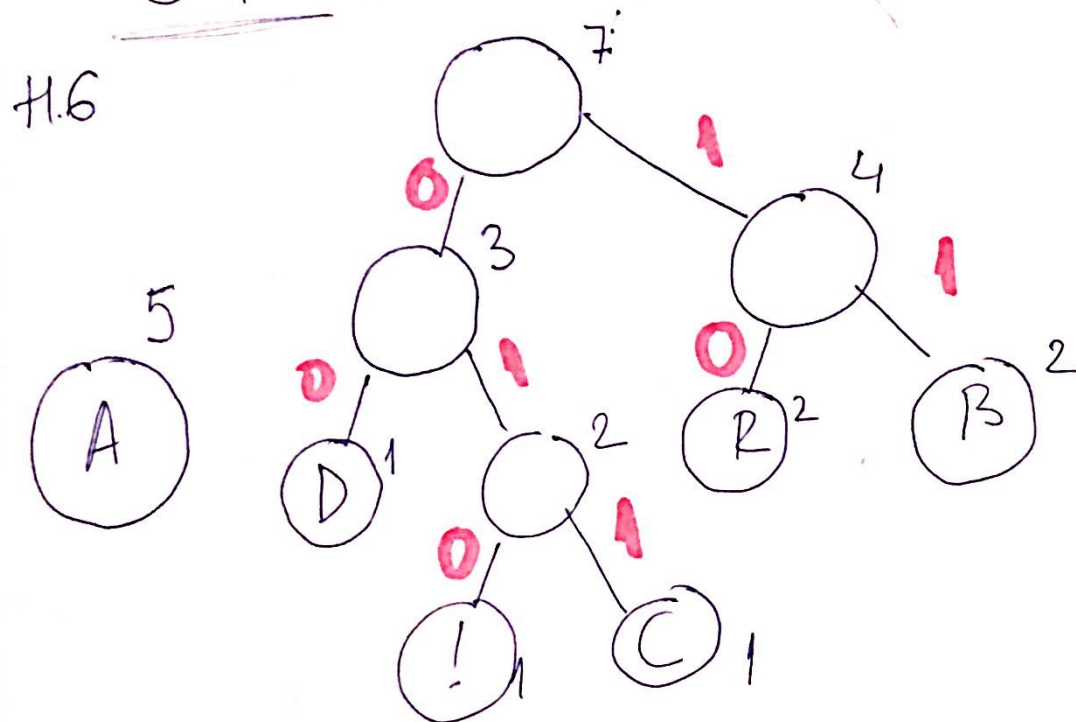
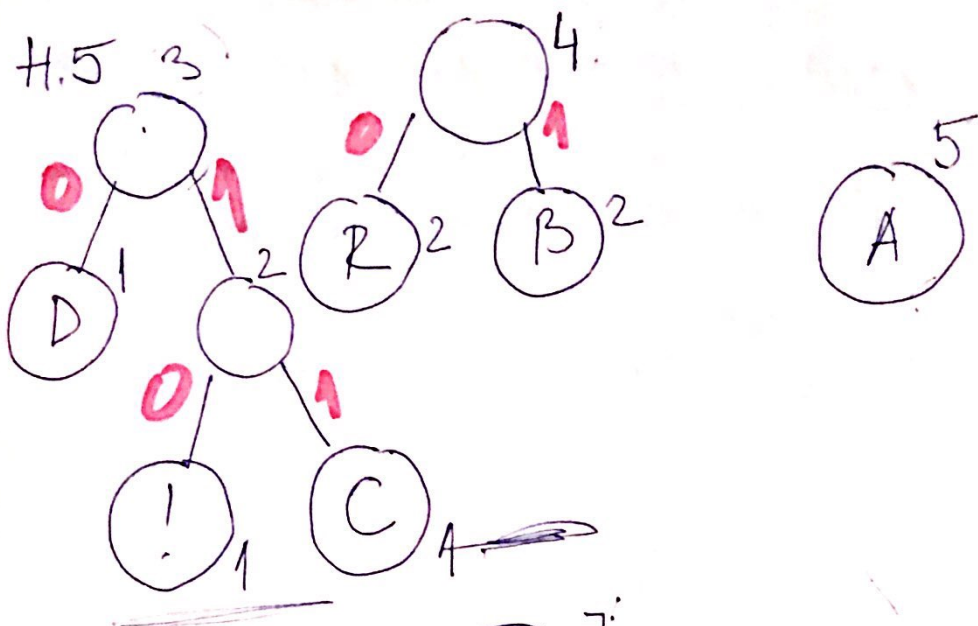


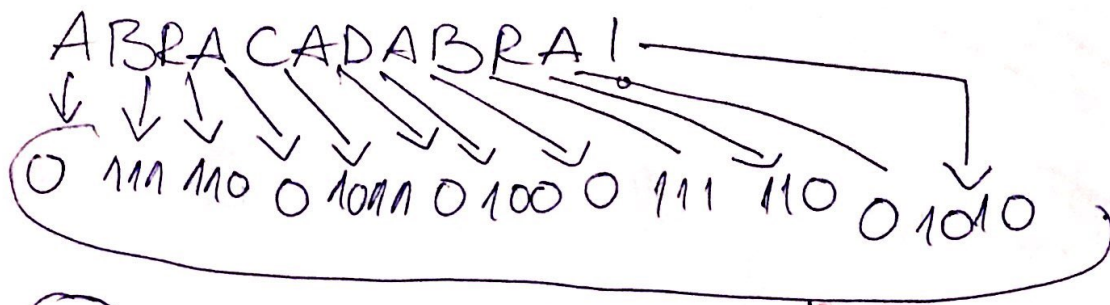
H.3



H.4



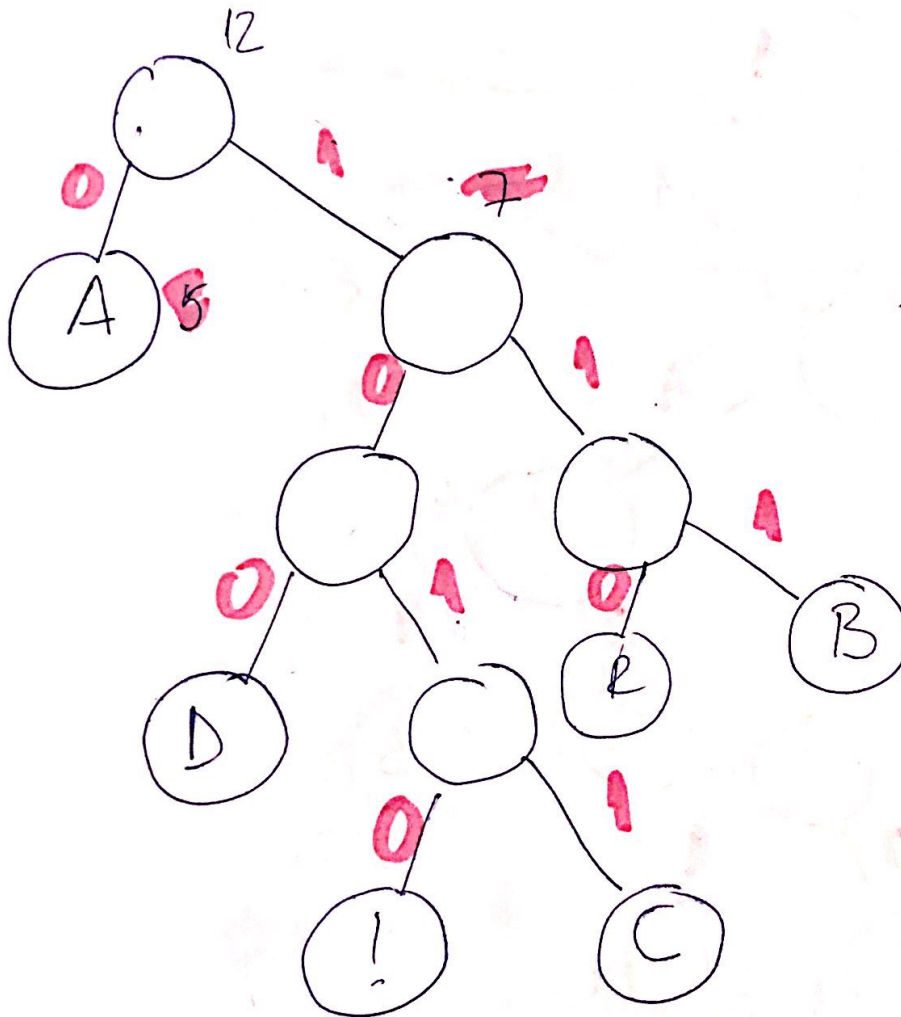




4) Dekompresimi.

A B R A C ~~A~~ ~~D~~ ~~A~~ B R A !

0111110 01011 ~~0100~~ ~~111~~ 1100 1010



A → 0
 B → 111
 C → 1011
 D → 100
 R → 110
 ! → 1010

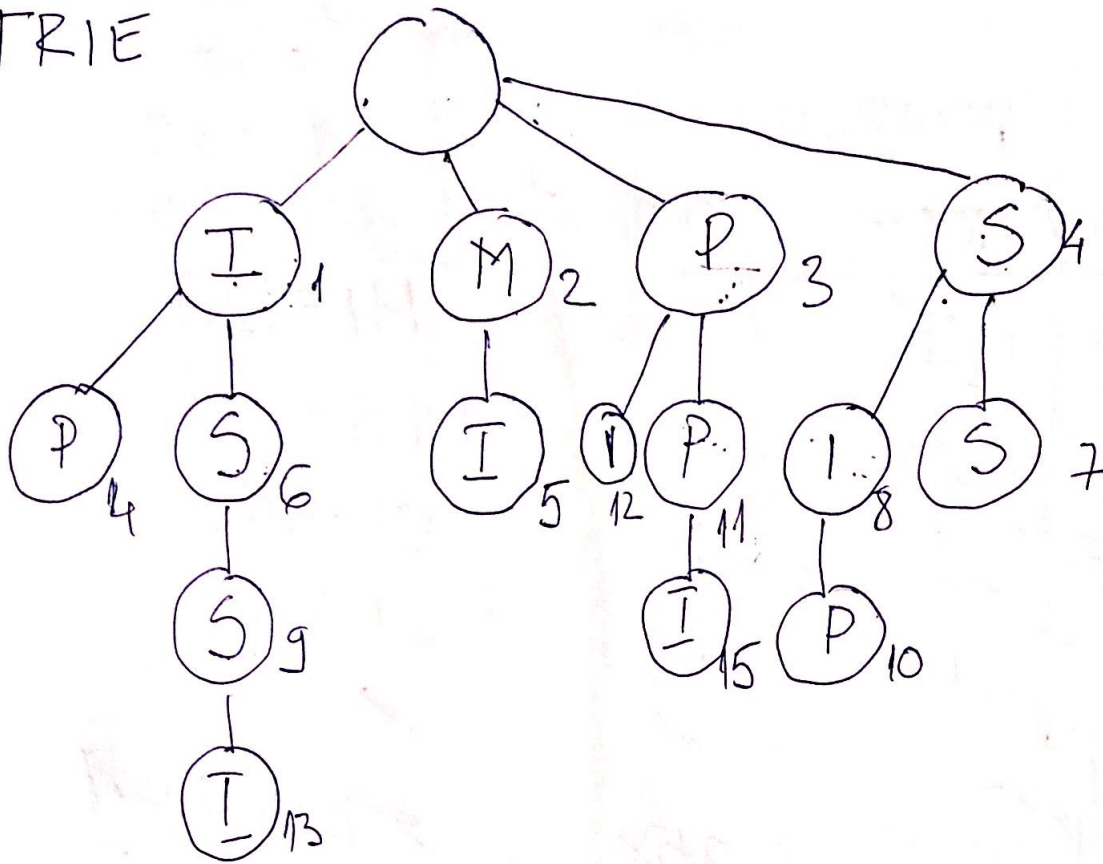
LZW

(*) MISSISSIPPISSIPPI

current	next	output	dictionary	
M	I	2	MI → 5	I → 1
I	S	1	IS → 6	M → 2
S	S	4	SS → 7	P → 3
S	I	4	SI → 8	S → 4
I	S	/	/	
IS	S	6	ISS → 9	
S	I	/	/	
SI	P	8	SIP → 10	
P	P	3	PP → 11	
P	I	3	PI → 12	
P	S	/	/	
IS	S	/	/	
ISS	I	9	ISSI → 13	
IS	P	1	ISP → 14	
IS	P	/	/	
IS	I	11	API → 15	
IS	I	1	/	

KOMPRESIMI=OUTPUT=2 1 4 4 6 8 3 3 9 1 11 1

TRIE



Dekompresimi

$\begin{matrix} \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ 2 & 1 & 4 & 4 & 6 & 8 & 3 & 3 & 9 & 1 & 11 & 1 \end{matrix}$

key	Value	g	ISS
1	I	10	SIP
2	M	11	PA
3	P	12	PI
4	S	13	ISSI
5	MI	14	IP
6	IS	15	PPI
7	SS		
8	SI		

MISSISSIPPI

(me ASCII)

Current	next	output	dictionary
M	I	77	MI → 257
I	S	73	IS → 258
S	S	83	SS → 259
S	I	83	SI → 260
I	S	/	/
IS	S	258	ISS → 261
S	I	/	/
SI	P	260	SIP → 262
P	P	80	PP → 263
P	I	80	PI → 264
PI	I	73	PII → 265
II	/	73	/

DICTIONARY

I → 73
M → 77
S → 83
P → 80

ASCII → 256

77 73 83 83 258 260 80 80 73 73
 ↓ ↓
 M I S S I S I P P I I